INGB413

QUALITY MANAGEMENT

INDIVIDUAL EXAM ASSIGNMENT

28 MAY 2025

TOTAL: 100

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Assignment Objective

The objective of this assignment is for you to demonstrate that you have mastered a fundamental understanding of quality management principles. Furthermore, this assignment assesses whether you have achieved the Engineering Council of South Africa's (ECSA) Graduate Attribute (GA) 4 at Exit Level.

ECSA Graduate Attribute 4: Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

General Instructions

For this individual assignment, you will have the opportunity to gather data during an experiment in the Learning Factory on 28 May 2025. After the data gathering, you need to submit a report on e Fundi (refer to instructions below) by **4 June 2025 @ 09:00**. This report will be assessed to determine whether you have successfully completed the assignment and achieved ECSA GA 4 at Exit Level.

Practical arrangements

You are assigned a specific time on 28 May 2025 when you can conduct your experiment in the Learning Factory, as communicated in eFundi. Data collection can only be done during this allocated time slot.

- Please arrive 10 minutes prior to your assigned time and wait outside the Learning Factory.
- Bring along a pen and paper to record the result (refer to the technology-free zone information stipulated below).

Occupational Health and Safety in the Learning Factory

The Learning Factory is a simulated factory environment, and therefore, relevant health and safety protocols should be adhered to.

The Learning Factory is a technology-free zone. No electronic devices (such as phones/tablets/laptops) will be allowed. There are lockers in the Learning Factory where you can lock up your personal belongings and keep the key with you for the duration of the practical experiment.

- Please wear closed shoes when entering the facility
- Each student will be provided with a safety vest associated with the production line to which you are assigned. Please wear this safety vest at all times.
- Please pay attention to all the safety instructions conveyed to you by the Learning Factory safety officer at the start of the designated session. This information will include details on safety exits and protocol to follow in case of an emergency or injury.

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Data Gathering Experiment

A tangram puzzle is a traditional Chinese puzzle that consists of seven geometric shapes that can be rearranged to form a variety of new shapes.

Tangra Inc. is a tangram manufacturing company that sources puzzle pieces from different suppliers, whereafter the pieces are assembled in a wooden block and then shipped to different customers.

The assembled tangram in the wooden block that is ready for packaging looks as follows:

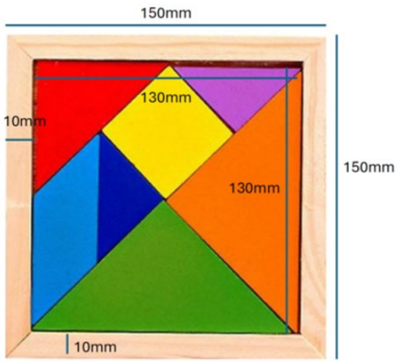

Figure 1 Assembled tangram ready for shipping

The quality management team specifies that they require the process (in terms of defective parts per million) to run at a 2-sigma level. *Tangra Inc*. has recently found an increase in defective puzzle pieces on its assembly lines. As a result, the assembly line is unable to produce enough tangrams to meet the current demand.

Tangra Inc. has approached you as an industrial engineer to investigate the problem and provide insights into the best way to identify defective puzzle pieces in their assembly line and ensure that as many tangrams as possible are assembled on the line. To do so, you have decided to conduct an experiment where two processes are tested.

Process 1

a. Receive raw material in bulk
b. Take samples of 5 and test for defects (sample size (n=5))
c. Remove all defective parts
d. Assemble tangrams

Test process 1 by following these process steps and assemble as many tangrams as possible in 15 minutes. Record all the defects, the number of completed tangrams as well as any other relevant information.

Process 2

a. Receive raw material already sorted by colour in lin bins at the assembly station
b. Pick parts from the lin bins
c. Start assembling a tangram
d. If you find any defects, discard the entire assembly (wooden board with puzzle pieces)
e. Count the number of puzzle pieces already placed and the number of defective pieces in each discarded assembly

Note: The number of pieces placed on the wooden board up until a defect is selected from the lin bin, is the sample size for that specific sample

Test process 2 by following these process steps and assemble as many tangrams as possible in 15 minutes. Record all the defects, the number of completed tangrams as well as any other relevant information.

Cost information

The following information regarding the cost of raw material and defects is known and shown in table 1:

Table 1 Tangra Inc costing information

Item	**Cost**
Puzzle piece	R2.00
Wooden board	R5.00
Defective semi-assembled item	R20.00

Each completed product can be sold for **R25.00**.

Report instructions

Write a report for the management team of *Tangra Inc*. where you highlight how you have investigated the data and what the findings of the two processes in terms of quality parameters are. Make use of quality management principles (which include statistical process control techniques) and provide insights on which of the two processes results in better quality outcomes and what the effect on profitability is. The process steps of both processes need to be clearly communicated to the management team.

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Ensure that you support the use of quality management techniques with relevant research and clearly state all assumptions. Also be sure to test relevant hypotheses, include valid control charts and use a relevant measure to comment on the capability of the process.

This is an individual assignment, and you are not permitted to work on the reports and data analysis with anyone else

Your report needs to be typed, and no handwritten or scanned reports will be accepted. Your report needs to consist of at least the following headings, but you are free to structure the report as you see fit.

i. Background
ii. Data collection and analysis
iii. Literature review
iv. Conclusion
v. References

There is no strict limit on the number of pages allowed for the report, but a general guideline is no more than 20 pages.

Submission information

Submit the report as a PDF document and your data separately as an Excel document on eFundi under the GA Assignment by **Wednesday 4 June 2025 @ 09:00**

Name your report as follows:

Name_student number_INGB413GA for example

Maria van Zyl_12345678_INGB413GA

Only individual assignments submitted via eFundi will be considered and no reports that are e-mailed will be accepted.

Important Information

When you submit your assignment, a Turnitin report will be generated that will be reviewed as part of your assessment. There is no minimum percentage of similarity required but the examiners will look for plagiarism issues in this report.

1. **WARNING:** Plagiarism of any kind is a serious offence and will result in severe steps in accordance with NWU policy.
2. Although it is permitted to make use of writing assistance tools such as Grammarly, you may not use large language AI models (i.e. ChatGPT) to write the report or analyse data.
3. You can only pass this module if you pass the GA assignment (above 50%) and your participation mark + GA Assignment mark is 50% or more.
4. No late assignments will be accepted for any reason.
5. No assignments will be accepted via e-mail
6. If you do not submit on eFundi on time you will fail the GA Assignment.